
J. Conor Schehl

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EDUCATION

Colorado School of Mines, Golden, CO | BS, Electrical Engineering | **GPA: 4.0** | **May 2028**

- **Relevant Coursework:** Circuits, Digital Logic, Feedback Control Systems, Signal Processing, Embedded Systems (upcoming), Active Devices (upcoming), VLSI (upcoming)

SKILLS

- **Hardware:** PCB Design (Altium), Schematic Capture, PCB Bring-up, STM32, SMD/SMT Soldering, Circuit Analysis, LTspice, JTAG, Oscilloscopes, DMMs, Hardware Debugging
- **Embedded:** Python, C/C++, CAN, SPI, I²C, UART, Linux, MATLAB, Digital Logic
- **General:** Microsoft Suite, Team Collaboration, Design Reviews, SolidWorks, DFR/DFM

EXPERIENCE

Mines Formula SAE | Electrical Engineer | **September 2025 - Present**

- Collaborated with a multidisciplinary team to make formula-style racecars perform competitively at the collegiate level.
- Designed and assembled PCBs and wire harnesses to acquire live telemetry data from the car's 20+ sensors to maximize performance while adhering to safety regulations.

Rosendin | Engineer Intern/Co-Op | **May 2026 - Present**

- Conducted thorough short circuit and arc flash analysis for power distribution systems ensuring strict protocol and safety compliance with NEC and IEEE codes.
- Planned system coordination and developed a disciplined approach to power management, fault isolation, and system reliability with ETAP and SKM.

PROJECTS

Analog to Digital Converter Data Acquisition & Power Distribution Board, Mines Formula SAE

- Architected a mixed-signal data acquisition board using an STM32 MCU and a dedicated 16-bit delta-sigma ADC communicating over SPI, implementing low-pass input filters for signal integrity.
- Designed a power distribution architecture to step a 12V battery down using a buck converter and LDO regulators to isolate analog and digital power rails.
- Engineered high-speed, impedance controlled differential pair traces for a 100BASE-TX Ethernet PHY for reliable, low latency communication with the Vehicle Control Unit.

Custom STM32 Microcontroller Development Board, Mines Formula SAE

- Designed a 4-layer development board for communication with Arduino Shields implementing serial communications like CAN, SPI, I²C, and UART for testing.
- Adhered to >80% power conversion efficiency (12V to 3.3V) using regulators isolating the switching converter from sensitive analog traces to minimize EMI.
- Optimized high-speed oscillator routing for minimizing EMI and maximizing clock stability.

Power and Safety-Critical Latching Board, Mines Formula SAE

- Designed a safety-critical analog board utilizing relays, optocouplers, diodes, fuses, and MOSFETs to provide reliable galvanic isolation, and safe EV shutdown.
- Validated fault detection logic via LTspice, and testing with an oscilloscope, DC power supply, and digital multimeter.
- Engineered HV/LV safety architecture to reduce the bill of materials and overall cost.